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References

Autism Spectrum Disorder as a Transdiagnostic Phenotype: Interface with Migraine, Functional Connectivity, and Sensory Gain

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Background and Aims

Autism spectrum disorder (ASD) is characterized by atypical sensory processing as a core pathophysiological feature, manifesting as hyper- or hyporeactivity to sensory stimuli. Contemporary models such as predictive coding propose disruptions in the hierarchical integration between top-down predictions and bottom-up sensory inputs, resulting in increased sensory gain. Similarly, migraine is characterized by sensory hypersensitivity, including photophobia and phonophobia, associated with cortical hyperexcitability and dysfunction of the trigeminovascular system. Evidence suggests that both conditions share alterations in thalamocortical circuits, salience networks, impaired habituation, and sensory amplification.

Objective

To analyze, from a neurobiological perspective, atypical sensory processing in ASD and its relationship with sensory manifestations observed in migraine, integrating findings from functional neuroimaging and contemporary theoretical models.

Methods

Narrative review based on databases such as PubMed, Scopus, and Web of Science databases, including studies published between 2015 and 2025. Priority was given to studies employing functional MRI (fMRI), connectivity analyses (seed-based approaches, graph theory, ALFF, ReHo), and investigations addressing predictive coding, cortical excitability, and the trigeminovascular system. Search terms included ASD, sensory processing, migraine, and functional connectivity.

Results

Findings demonstrate convergence across multiple neurobiological levels. Increased sensory gain and deficits in predictive coding were observed, characterized by greater reliance on afferent inputs and reduced top-down modulation, leading to hyperresponsivity. Cortical hyperexcitability is present in both conditions; in migraine, it is associated with cortical spreading depression, particularly in visual areas and correlated with photophobia, whereas in ASD it is observed in primary sensory cortices. Thalamocortical dysfunction impairs sensory gating, contributing to perceptual overload. Altered activity in the anterior insula and anterior cingulate cortex affects salience attribution, enhancing aversive responses. In migraine, activation of the trigeminovascular system contributes to amplification of non-nociceptive stimuli, suggesting interaction with sensory networks. Additionally, both conditions exhibit impaired habituation and maladaptive neuroplasticity.

Conclusions

Atypical sensory processing in ASD may be conceptualized within a spectrum of sensory dysfunction that includes migraine-related phenomena such as photophobia and phonophobia. The convergence of predictive coding deficits, increased sensory gain, thalamocortical dysfunction, and cortical hyperexcitability supports a shared neurobiological substrate. Integration of fMRI findings with advanced theoretical models enhances transdiagnostic understanding and may inform therapeutic strategies targeting neural connectivity and excitability.